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ADELAIDE GEOSYNCLINE AND STUART SHELF

SUNDRY MINERALISATION

BY

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INTRODUCTION

The marked clustering of occurrences of base metal and gold mineralisation in outcropping Precambrian and Cambrian rocks in the eastern and central portion of South Australia (Figure 1) clearly outlines the Adelaide Geosyncline and the bordering Stuart Shelf as a major metallogenetic province (Dickinson & Sprigg, 1953; Thomson, 1965). Copper is the major metal while gold, lead, zinc and silver are widely distributed but have proved to be of minor importance.

Although silver-bearing lead ores were discovered at Glen Osmond, and copper at Montacute, in 1841, it was not until copper was discovered at Kapunda in 1842 that widespread interest in metal mining was aroused in South Australia. Output of copper ore at Kapunda in the period 1842 to 1878 comprised 70 000 tonnes which yielded 13 000 tonnes of metal. Ore production at Burra during the years 1844 to 1877, amounted to 700 000 tonnes from which 71 600 tonnes of copper were recovered. Blinman, the largest and most productive copper mine in the Flinders Ranges, was operated during the period 1861 to 1918 when 200 000 tonnes of ore were raised to yield 9 700 tonnes of copper metal.

Throughout the State, over 800 named copper and 200 lead mines and prospects have been recorded and at least 80% of these are located within this province (Brown, 1908). Until the end of World War I approximately 420 000 tonnes of copper ore were produced, of which the Wallaroo and

Moonta Mines contributed 333 600 tonnes; most of the remainder was derived from the area under review.

Smelters were constructed at a number of centres including Mount Fitton South, Yudnamutana, Mount Rose, Mountain of Light, Sliding Rock, Blinman, Prince Alfred, Spring Creek, Burra, Kapunda, Kanmantoo and Mount Gunson but these were mostly short-lived as ore reserves generally proved to be insufficient to sustain their operation. Mining was based on near-surface secondarily enriched carbonate ores and almost without exception, the primary sulphide zones proved to be too narrow and of too low grade to be recoverable.

Gold discoveries at Echunga in 1852, followed by those at Barossa (1868), Waukaringa (1872), Woodside (1882), Mannahill (1885), Teetulpa (1886), Wadnaminga (1888) and Mongolata (1930) resulted in comparatively small production. Output of silver, lead and zinc have been unimportant.

During the period 1920 to 1960 exploration for and production of base metals were virtually at a stand still though 1 100 tonnes of copper were recovered at the Broken Hill Associated Smelters at Port Pirie from 32 000 tonnes of ore mined at Mount Gunson during World War II. Few new deposits were discovered during this time and exploration was generally confined to sporadic and discouraging investigations by a few exploration groups in areas of previous mining activity.

Active exploration was resumed throughout the region

in the mid 1960's to prove extensions to mineralised zones at a number of old mining centres. Geological investigations have been supported by geochemistry as an exploration method and by geophysics, particularly induced polarisation techniques; follow-up drilling has established reserves of recoverable sulphide ore at Kanmantoo, of oxidized copper ores at Burra and at Mount Gunson and of oxidized zinc ores at Puttapa and Aroona. Elsewhere, significant concentrations of copper, lead and zinc have been proven.

MINERALISATION CONTROLS

Figure 1 depicts the major faults and shears and the relation of known copper lead/zinc and gold mineralisation within Precambrian and Cambrian rocks of the Adelaide Geosyncline and Stuart Shelf. It is apparent that the density of regional fracturing and the distribution of mineralisation are related.

In the northern Flinders Ranges the major fault lineaments trend NE-SW and NW-SE, conforming to the trend of the Paralana/Anabarna and Nor'West Fault systems and confined by fundamental north-south trending systems, to form a regular network of intersecting faults and shears. These intersections, where imposed on structural culminations, controlled the emplacement of diapiric breccias which are the loci of copper and barium mineralisation through-out the Flinders Ranges. Southwards the major faults are generally directed meridionally and are arcuate

to the southwest in conformity with the prolongation of Fleurieu Peninsula to Kangaroo Island.

Orogenic movements which were initiated in the Cambrian culminated in Ordovician igneous activity along the eastern and southern margins of the belt. This is considered (Thomson, 1969) to have been the metallogenic event responsible for the introduction of numerous small metallic ore deposits at many stratigraphic levels throughout the Adelaide Geosyncline. The pattern of major faults and shears, some of which have been intermittently active since the Proterozoic, dominate the geosynclinal zone throughout its extent and is the major control in localisation of base metal and gold mineralisation. Unconformities have also played a primary role in controlling mineral deposition not only in the Adelaide Geosyncline but also on the Stuart Shelf.

A zonation of lead/zinc, barytes, uranium and gold is evident whereas copper is distributed throughout.

MINERALISATION IN THE ADELAIDE GEOSYNCLINE

Willouran sediments, whether in stratigraphic context or exposed by emplacement in diapiric cores, are hosts to anomalous concentrations of copper over wide areas. Where they unconformably overlap crystalline basement rocks of the Mount Painter Complex the Lower Callanna Beds and succeeding units in the Yudnamutana area are hosts to numerous occurrences of copper which were worked intermittently during the period 1862 to 1920. These comprise defined lodes in fissure veins, stockworks and replacements

in a restricted stratigraphic zone, penetrated by faulting and in close proximity to the Mudnawatana Granite (Blissett, 1971).

Conglomerates, sandstones, siltstones and shales, comprising units of the Upper Callanna Beds mark a second transgressive phase in which there are concentrations of oxidized copper minerals, traceable from O'Donoghue's Castle Mine, northwards to the Arkaroola locality, round the basement core beyond the Blue Mine to Yudnamutana.

It is apparent that there is a close association of copper with centres of diapiric intrusion throughout the Flinders Ranges (Coats, 1964) and most mines are situated within these structures, in adjacent rim rocks or along faults which radiate from them. The diapirs result from the breaking out of Willouran dolomites, shales, siltstones and sandstones from their confining environment and their intrusion by plastic flowage and injection into the overlying sedimentary cover in favourable structural situations. Basic igneous intrusions are generally associated with diapirism and these appear to have played a role in introduction of copper. Within the Oraparinna Diapir a sequence of layered melaphyres and basaltic lavas which are probable correlatives of the Wooltana Volcanics include copper sulphides as amygdule infillings (Johns, 1971). An association of copper with dolerite is apparent at Mount Lyndhurst, Mount Burr, Booleroo, Mount Coffin, Mountain of Light,

Beltana, Loch Ness, Mucatoona, Blinman, Oraparinna, Enorama, Arkaba, Worumba, Spalding and Burra.

Within the diapiric structures, 'rafts' of intensely folded Willouran rocks are separated and disrupted by masses of breccia and, often, structureless dolomite. Disseminated sulphides including chalcopyrite, of apparent stratiform habit have been recognised in most of the diapirs within dismembered dolomites that are similar to those of the Wywyana Formation and in siltstones and shales comparable in lithology with certain units of the River Wakefield Group. Copper contents up to 1% have been determined and are traceable in some cases along strike for several kilometres within these structures.

Blinman Mine was based on a tabular, oxidized ore body enclosed by a raft of dolomite within the Blinman Diapir. The lode ranged from 10 ft. to 50 ft. in width and was 350 ft. long and workings extend to a depth of 600 ft. Copper contents ranged up to 30% (Dickinson, 1942).

Rocks of the Yudnamutana Sub-group which rest unconformably on 'islands' of Willouran rocks, peripheral to the Mount Lyndhurst, Mount Burr, Mount Coffin, Boolooroo and Mountain of Light culminations contain copper mineralisation, traceable within a narrow stratigraphic interval of about 50 metres, discontinuously over strike lengths of several tens of kilometres. It is evident that copper, derived from primary mineralisation within the diapirs, has been redistributed by groundwater circulation and facilitated by the brecciated host rocks,

adjacent to the unconformity. The White Lead, Paull's Consolidated, Victory, Elsie Adair, Mount Coffin and Diamond Jubilee mines were based on concentrations of copper carbonate of this type. Drilling at these centres has demonstrated that irregular concentrations of malachite, azurite and chalcocite occur above the water table along bedding planes and cross cutting fractures within shales, siltstones and tillitic sandstones.

The Prince Alfred and Red Hill mines disclose copper mineralisation in the lower part of the Tapley Hill Formation in proximity to an unconformity.

In the southern Flinders Ranges and in the Mount Lofty Ranges there are a number of previously worked copper mines in which mineralisation has proven to be weak, wall-rock alteration slight and the narrow low grade siliceous veins emplaced along small faults, shear zones and cleavage fractures. Leaching and secondary enrichment resulted in the accumulation of small high grade deposits which have been formerly mined at a number of widely scattered centres including Paratoo, Spring Creek, Baroota, Mount Remarkable, Spalding, Stanley, Utica, Burra, Apoinga, St. Elmo, Karkulto, Princess Royal, Montacute and Almanda.

The Kapunda copper ore bodies are enclosed in Sturtian kaolinitic siltstones at the crest of an anticline. Thirty discrete veins consisting of carbonates, oxides and secondary sulphides in quartz were mined with some of the enclosing country rock in the zone of secondary enrichment and in primary sulphides to a depth of 430 ft.

below the surface. Recent drilling operations have established several million tons of remnant oxidized copper ores averaging 1% copper in the old mine area.

Fault zones within Lower Cambrian carbonate hosts have been the loci of copper mineralisation at a number of centres in the northern Flinders Ranges including Sliding Rock, Red Range, Ajax, Copper King, Mount Bayley, Angepena, Mount Chambers, Radford Creek and Warrakimbo. The most important of these was Sliding Rock and during the period 1871 to 1908 some 1 750 tonnes of rich carbonate ore and native copper were recovered from above the water table. At Mount Arden, Kanyaka, Moro and Balcanoona shallow workings were based on apparently stratiform copper carbonates within the Parachilna Formation.

Significant concentrations of lead and zinc with minor copper have proven to be widely distributed within carbonate rocks of the Lower Cambrian (Johns, 1972a). These deposits may be classified as Mississippi Valley type. Oxidized lead and zinc mineralisation, derived from stratiform sulphides, have been concentrated by circulating groundwaters in and adjacent to major fault zones.

Galena occurs as disseminations and in veinlets in basal dolomites in proximity to the boundary fault which determines the western margin of the Ediacara Basin and in the matrix of leached collapse breccias peripheral to the basin (Binks, 1972). During the

period 1888 to 1913 about 24 000 tonnes of ore averaging 31% lead and 8 ozs. of silver per tonne were mined from shallow pits and underground galleries. Drilling has outlined reserves of 30 million tonnes over an average thickness of 20 metres with an average grade of 0.9% lead (Nixon, 1967).

Galena has been mined on a small scale at Wirrealpa, Flinders, Fountain Head, Mount Lyall, Mount Chambers, Parachilna and Rapid Bay. Exploration has disclosed base metal sulphides within Lower Cambrian carbonates along the eastern and western margins of the Flinders Ranges, the Torrens "hinge zone" and adjacent to the coast between Sellick Hill and Rapid Bay.

Oxidized near-surface secondary concentrations of willemite with associated lead minerals and manganese and iron oxides have been located at Puttapa, Aroona and at Third Plain. Reserves of about one million tonnes, ranging up to 52% zinc have been established by drilling at Puttapa.

Recent exploration has revealed the presence of copper-molybdenite mineralisation in association with Ordovician granites at Anabama, Netley and Bendigo adjacent to the Anabama Fault. A number of features, including brecciation, hydrothermal alteration and type of mineralisation invite comparisons with the porphyry copper type.

The several occurrences of known lead/zinc mineralisation in Adelaidean rocks have yielded quite small production. Thin cross cutting tabular veins have been exploited at Appeallinna Well, Oodnapanakan Hut, Ooloo, Constitution Hill, Mallee Hut and Gilead P. Beck prospects. Several hundred tons of galena have been produced from four parallel lodes, one metre in width, at the Avondale Mine and sphalerite was encountered below the water table. About 1 500 tonnes of 30% zinc ore were won from the Billy Springs Mine, chiefly in the form of smithsonite which was the principal ore in a complex of zinc, lead and copper minerals. There has been desultory production of galena from the Baratta-Eukaby field.

In the Olary area several unimportant occurrences of galena are known at Trinkaleena, Wadnaminga, Vranes, Lady Elizabeth, Uncle Tom and Winckler's prospects.

In the Mount Lofty Ranges the principal, now defunct, lead mines include the Glen Osmond group, Morialta, Hahndorf and Almanda.

Gold is rare in the northern Flinders Ranges. Small production has been derived from deposits located within or adjacent to diapirs at Boolooroo, Mount Ogilvie, Angepena, Patawarta, and Mattwarangula and it is a minor associate of some copper ores. In the Olary region gold-bearing quartz reefs and stock-works have yielded irregular values at King's Bluff, Nillinghoo, Wadnaminga, Teetulpa, Mannahill and Kirkeeks Treasure. Extensive

selective mining has been undertaken at the Waukaringa Mine to yield an average battery return of 17 dwts. of gold per tonne during the years 1872 to 1889.

The Mount Grainger field produced 3 600 ozs. of gold averaging almost 10 dwts. per tonne from quartz veins in kaolinised tillite in a structurally complex zone. At Mongolata gold occurs in quartz-limonite veins in sandstone; between 1930 and 1940 approximately 7 500 tonnes of ore were treated for the recovery of 11 000 ozs. of gold. The workings extend over a strike length of three kilometres and attain 85 m in depth. In the same general area relatively unimportance occurrences of gold have been mined in the Pitcairn Range, at Minjibbie, Mittopitta and Ulooloo.

In the Mount Lofty Ranges auriferous quartz reefs within Adelaidean Rocks are rare and the Tertiary placers appear to have been derived principally from erosion of primary deposits within the crystalline basement.

MINERALISATION ON THE STUART SHELF

Copper deposits were discovered adjacent to the western shore of Pernatty Lagoon in 1875 and desultory mining operations resulted in the production of 3 250 tonnes of hand-dressed ore, generally containing 8% to 16% copper in 1937. During the period 1941 to 1943 a further 32 000 tonnes of ore were mined

for the production of 1 100 tonnes of copper metal and 14 523 ozs. of silver. Renewal of exploration activity in 1965 led to the re-opening of the mine by Mount Gunson Pty. Ltd. in 1970 when open-cut reserves were reported to total three million tonnes of +1% copper ore. The latest phase of mining was based on the Gunyah deposits, underlying the floor of an arm of Pernatty Lagoon, and chalcocite was separated from the sandstone-quartzite host rocks by flotation in a mill constructed on the adjacent tableland.

The known copper deposits comprise discrete, shallow, more or less tabular bodies of generally low grade that range up to 5 m in thickness and are concentrated along unconformities which separate the Whyalla Sandstone and Woocalla Dolomite from the underlying Pandurra Formation. Copper minerals which include chalcocite, bornite, covellite, malachite, atacamite and chrysocolla occur as a cement to rounded quartz grains in the friable Whyalla Sandstone and as infillings of irregular fractures in dense quartzite on the surface of the unconformity. Sulphides (principally chalcocite) occur in the lagoon deposits and in the deeper levels of the tableland workings while carbonates and silicates are the common ores of copper in outcrop and where exposed in the shallow tableland deposits.

Copper also occurs within the Woocalla Dolomite, occasionally in association with barytes, fluorite

and manganese at Mount Gunson in the Dolomite, Sweet Nell and Fair Nell prospects.

Geological features of the Pernatty Lagoon-Mount Gunson area have been previously reported by Dickinson (1942) and Johns (1968). The results of recent exploration undertaken in the region by Noranda Australia Ltd. have been reported by Johns (1972b).

The Stuart Shelf comprises an underformed cratonic cover of Adelaidean shallow water marine sediments which rest on metamorphic and igneous rocks of the Gawler Craton. Pyroxene basalts of presumed Willouran age and exceeding 200 m in thickness are succeeded by the Pandurra Formation, a clastic sequence of horizontally disposed quartzites at least 200 m in thickness, that outcrops as an erosional high in the "Pernatty Culmination". Erosional irregularities of the surface controlled the sedimentation of the Woocalla Dolomite which rests unconformably on that surface immediately to the west and south of Pernatty Lagoon. The dolomite which ranges up to 150 m in thickness outcrops along the western shore of Pernatty Lagoon. It exhibits features of shallow water deposition; intraformational breccias and crenulated bedding with abrupt terminations are suggestive of a stromatolite reef. Drilling has indicated irregularities on the surface of the Pandurra Formation in which finely laminated dolomites and shales have been deposited. It is inferred

that these accumulated in a restricted lagoonal environment with marginal reef complexes about basement highs which correspond to the Pernatty Culmination.

The flat lying Tent Hill Formation is typically exposed in tableland margins throughout the extent of the Arcoona Plateau. The lowermost Member, the Whyalla Sandstone, rests disconformably on the Woocalla Dolomite or, where this was eroded, directly on the Pandurra Formation. The succeeding members include Woomera Shale, Corraberra Sandstone and Arcoona Quartzite.

Copper mineralisation appears to be concentrated in the Whyalla Sandstone in the Pernatty Lagoon locality adjacent to the stromatolite reef complex and marginal to the Pernatty Culmination. Zinc and lead, in subeconomic concentrations, have been proven by drilling to occur over a wide area in dolomitic black shales. Mineralisation consists of disseminated aggregates of pyrite, sphalerite, galena and chalcopyrite ranging to 1 mm diameter. Several zones ranging up to 10 m in thickness have been proven to contain 0.2% lead and 0.3% zinc with some sections 2 m in thickness containing in excess of 1.0% combined sulphides.

Though no economic concentrations of lead or zinc have disclosed the existence of geochemically anomalous levels of base metal mineralisation have been established in a stratiform setting over a wide area. The localisation of mineralisation is considered to have been governed

by several possible controls including the proximity of the Pernatty Culmination, a positive structural feature which had a profound influence on sedimentation and the unconformities separating reef dolomite, a black shale facies and sandstones.

SUMMARY

It is considered that widespread base metal deposits of the Adelaide Geosyncline were emplaced during the Ordovician. Copper is the major metal with lead, zinc and gold of lesser importance.

Structural control has been of paramount importance in localising diapiric emplacement and dolerite intrusion and in providing channel ways for circulation of mineralising fluids in proximity to intrusive centres and the mobilisation and concentration of deep seated mineralisation.

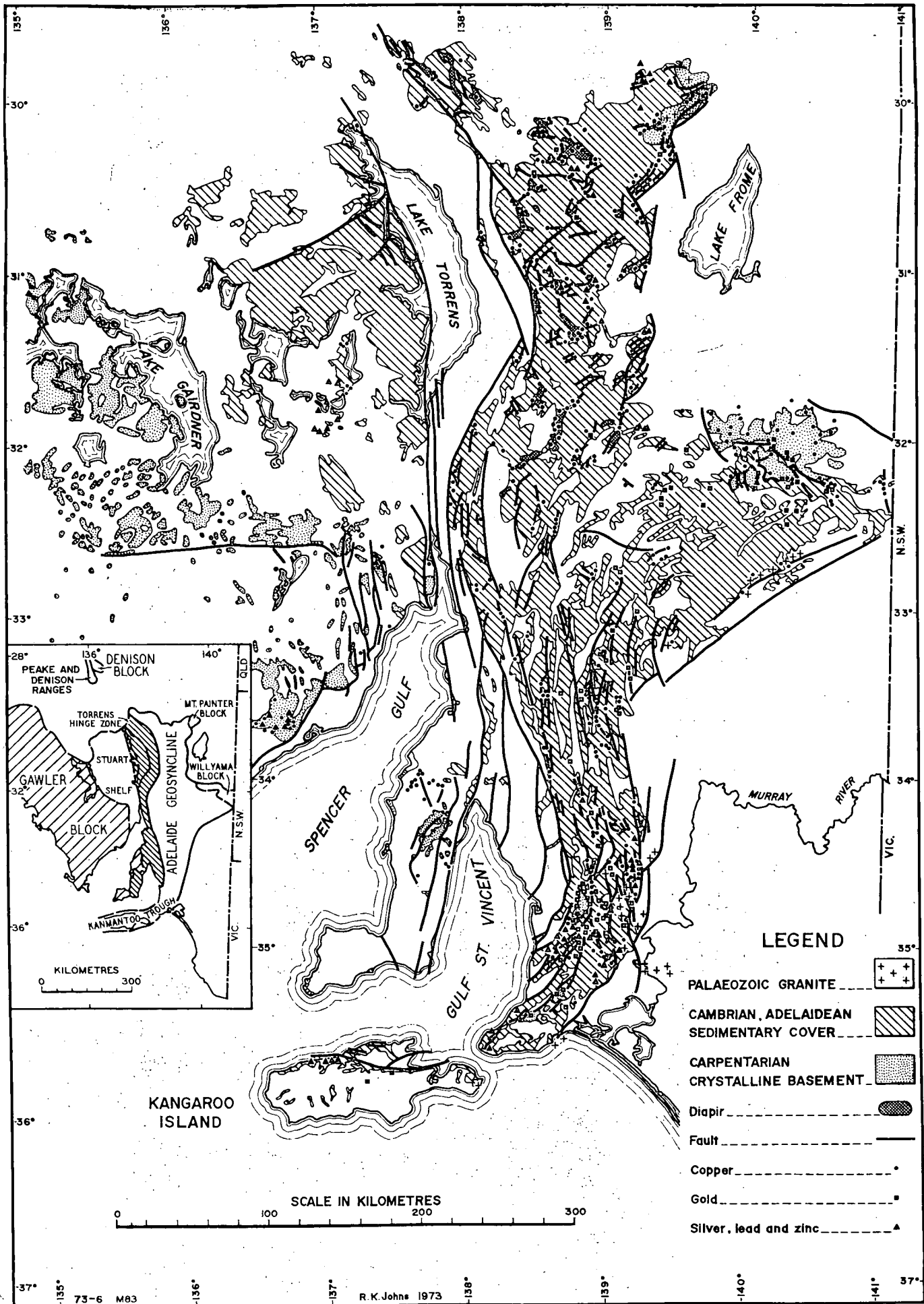
The role of unconformities in ore localisation within the geosyncline and on the Stuart Shelf is stressed since they provided favourable loci for introduced mineralisation and potential aquifers for ore fluids.

Deep weathering and prolonged subaerial erosion on base levelled surfaces during the Cainozoic era resulted in the leaching and redistribution of earlier deposits and favoured the accumulation of high grade secondarily enriched mineral deposits.

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